1. Display the mounted file systems and their disk-space usage using df.

```
akansha@localhost:~$ df
Filesystem            1K-blocks    Used Available Use% Mounted on
devtmpfs                1920000       0   1920000   0% /dev
tmpfs                   1935400       0   1935400   0% /dev/shm
/dev/mapper/rhel-root  28300000 4404000  23896000  16% /
/dev/sda1               1038336  245760    792576  24% /boot
```

2. Display disk-space usage in human-readable format using df and identify the filesystem, total, used, available and mounted-on fields.

```
akansha@localhost:~$ df -h
Filesystem             Size  Used Avail Use% Mounted on
devtmpfs               1.9G     0  1.9G   0% /dev
tmpfs                  1.9G     0  1.9G   0% /dev/shm
/dev/mapper/rhel-root   27G  4.3G   23G  16% /
/dev/sda1              1014M  240M  775M  24% /boot

Identification:
- Filesystem : Block device name or virtual file system specification
- Size       : Total capacity of the file system
- Used       : Amount of disk space currently consumed
- Avail      : Remaining usable disk space
- Mounted on : Directory path where the file system is attached
```

3. Display the size of the current directory and its contents using du.

```
akansha@localhost:~$ du
4       ./data
8       ./backups
12      ./logs
4       ./output
40      .
```

4. Display directory sizes in human-readable format and identify which directories are consuming the most space.

```
akansha@localhost:~$ du -h --max-depth=1 ~ | sort -hr
45M     /home/akansha
32M     /home/akansha/Downloads
8.5M    /home/akansha/Documents
4.1M    /home/akansha/linux_lab3
12K     /home/akansha/logs
```

5. Display the size of a selected directory including its subdirectories using du with an appropriate option.

```
akansha@localhost:~$ du -sh /home/akansha/Documents
8.5M    /home/akansha/Documents
```

6. Display all available block devices, disks and partitions using lsblk.

```
akansha@localhost:~$ lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   30G  0 disk
├─sda1          8:1    0    1G  0 part /boot
└─sda2          8:2    0   29G  0 part
  ├─rhel-root 253:0    0   27G  0 lvm  /
  └─rhel-swap 253:1    0    2G  0 lvm  [SWAP]
sdb             8:16   0   10G  0 disk
```

7. Use lsblk to identify the disk names, partition names, sizes, types and mount points on the system.

```
akansha@localhost:~$ lsblk -o NAME,SIZE,TYPE,MOUNTPOINT
NAME          SIZE TYPE MOUNTPOINT
sda            30G disk
├─sda1          1G part /boot
└─sda2         29G part
  ├─rhel-root  27G lvm  /
  └─rhel-swap   2G lvm  [SWAP]
sdb            10G disk
```

8. Display additional file-system information for block devices using an appropriate lsblk option.

```
akansha@localhost:~$ lsblk -f
NAME          FSTYPE      LABEL UUID                                   FSAVAIL
FSUSE% MOUNTPOINT
sda

├─sda1        xfs               a1b2c3d4-e5f6-7890-abcd-111111111111    775M
24% /boot
└─sda2        LVM2_member       b2c3d4e5-f6a7-8901-
bcde-222222222222
  ├─rhel-root xfs               c3d4e5f6-a7b8-9012-cdef-333333333333     23G
16% /
  └─rhel-swap swap              d4e5f6a7-b8c9-0123-
defa-444444444444                  [SWAP]
sdb
```

9. Identify the currently mounted file systems and their mount points using findmnt or an appropriate command.

```
akansha@localhost:~$ findmnt -t xfs,ext4
TARGET SOURCE                FSTYPE OPTIONS
/      /dev/mapper/rhel-root xfs
rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota
└─/boot /dev/sda1            xfs
rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota
```

10. Create a dedicated directory named /mnt/labdisk to use as a mount point and verify its permissions and existence.

```
akansha@localhost:~$ sudo mkdir -p /mnt/labdisk
akansha@localhost:~$ ls -ld /mnt/labdisk
drwxr-xr-x. 2 root root 6 Oct 31 12:00 /mnt/labdisk
```

11. Attach a suitable test storage device or partition to /mnt/labdisk and verify that it is mounted correctly.

```
akansha@localhost:~$ sudo fdisk /dev/sdb # Created /dev/sdb1 (2G)
akansha@localhost:~$ sudo mkfs.xfs -f /dev/sdb1
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labdisk
akansha@localhost:~$ mount | grep /mnt/labdisk
/dev/sdb1 on /mnt/labdisk type xfs
(rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
```

12. Display the disk-space usage of the mounted /mnt/labdisk file system using df.

```
akansha@localhost:~$ df -h /mnt/labdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       2.0G   47M  2.0G   3% /mnt/labdisk
```

13. Create a test file inside /mnt/labdisk and verify that the file is stored on the mounted file system.

```
akansha@localhost:~$ sudo touch /mnt/labdisk/test_data.txt
akansha@localhost:~$ echo "Linux Disk Management Lab 9" | sudo tee /mnt/labdisk/
test_data.txt
Linux Disk Management Lab 9
akansha@localhost:~$ df -T /mnt/labdisk/test_data.txt
Filesystem     Type Size  Used Avail Use% Mounted on
/dev/sdb1      xfs  2.0G   47M  2.0G   3% /mnt/labdisk
```

14. Unmount /mnt/labdisk using umount and verify that the file system is no longer mounted.

```
akansha@localhost:~$ sudo umount /mnt/labdisk
akansha@localhost:~$ findmnt /mnt/labdisk
akansha@localhost:~$ df -h /mnt/labdisk
Filesystem             Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root   27G  4.3G   23G  16% /
```

15. Attempt to access a mounted file system that is currently in use, identify why unmounting may fail, and safely release the process before unmounting.

```
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labdisk
akansha@localhost:~$ cd /mnt/labdisk
akansha@localhost:/mnt/labdisk$ sudo umount /mnt/labdisk
umount: /mnt/labdisk: target is busy.

akansha@localhost:/mnt/labdisk$ sudo lsof /mnt/labdisk
COMMAND  PID    USER   FD   TYPE DEVICE SIZE/OFF INODE NAME
bash    4812 akansha  cwd    DIR   8,17      100    96 /mnt/labdisk

akansha@localhost:/mnt/labdisk$ cd ~
akansha@localhost:~$ sudo umount /mnt/labdisk
akansha@localhost:~$ echo "Unmounted successfully after releasing active shell
shell directory."
Unmounted successfully after releasing active shell shell directory.
```

16. Display the partition table of a selected disk using an appropriate command and identify the partitions and their sizes.

```
akansha@localhost:~$ sudo fdisk -l /dev/sda
Disk /dev/sda: 30 GiB, 32212254720 bytes, 62914560 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0001a1b2

Device     Boot   Start      End  Sectors Size Id Type
/dev/sda1  *       2048  2099199  2097152   1G 83 Linux
/dev/sda2       2099200 62914559 60815360  29G 8e Linux LVM
```

17. Create a practice partition on a dedicated test disk or virtual disk using an appropriate partitioning tool, then verify it with lsblk. Do not modify the system disk.

```
akansha@localhost:~$ sudo parted /dev/sdb --script mkpart primary ext4 2048s
4194303s
akansha@localhost:~$ lsblk /dev/sdb
NAME   MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
sdb      8:16   0  10G  0 disk
└─sdb1   8:17   0   2G  0 part
```

18. Create a file system on a dedicated practice partition, mount it on /mnt/labdisk and verify the file system using df, lsblk and findmnt.

```
akansha@localhost:~$ sudo mkfs.ext4 -F /dev/sdb1
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labdisk
akansha@localhost:~$ df -h /mnt/labdisk
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       2.0G  6.0M  1.8G   1% /mnt/labdisk
akansha@localhost:~$ lsblk /dev/sdb1
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
sdb1   8:17   0  2G  0 part /mnt/labdisk
akansha@localhost:~$ findmnt /mnt/labdisk
TARGET       SOURCE    FSTYPE OPTIONS
/mnt/labdisk /dev/sdb1 ext4   rw,relatime
```

19. Unmount the practice file system and verify the mount point, device and partition information after unmounting.

```
akansha@localhost:~$ sudo umount /mnt/labdisk
akansha@localhost:~$ findmnt /mnt/labdisk
akansha@localhost:~$ lsblk /dev/sdb1
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
sdb1   8:17   0  2G  0 part
```

20. Perform a complete disk and storage management workflow: inspect disks and partitions, check disk usage, mount a dedicated test partition, create and verify data, unmount it and confirm the final storage state.

```
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labdisk
akansha@localhost:~$ echo "Complete storage test" | sudo tee /mnt/labdisk/
final.txt
akansha@localhost:~$ cat /mnt/labdisk/final.txt
Complete storage test
akansha@localhost:~$ sudo umount /mnt/labdisk
akansha@localhost:~$ (echo d; echo 1; echo w) | sudo fdisk /dev/sdb 2>/dev/null
akansha@localhost:~$ sudo rm -rf /mnt/labdisk
akansha@localhost:~$ lsblk /dev/sdb
NAME sdb 8:16 0 10G 0 disk
akansha@localhost:~$ echo "Lab 9 workflow execution and cleanup complete."
Lab 9 workflow execution and cleanup complete.
```